



CLINICAL PROGRAMMING BOOTCAMP • MODULE 04

Reading Raw Clinical Data

Before you can map anything, you have to get the data in — with the right types, the right dates, and your eyes open for what's wrong with it.

Hands-on: Notebook 03 • Importing Raw Data (SAS & R)

By the end of this module you can...

1

Import a raw CSV correctly in both SAS and R, with the types you intended.

2

Recognise the four data types character, numeric, date, and missing — and tell them apart.

3

Spot the classic traps leading zeros, mixed date formats, blanks, wide structures.

4

Inspect before you map run a standard checklist on any unfamiliar dataset.

5

Plan a mapping list exactly what must change to reach SDTM.

What a raw extract actually looks like

One file per CRF form — note Demographics and Disposition are separate forms, completed at different times.

File	CRF form	Structure	Rows
dm_raw.csv	Demographics (screening)	one row per subject	8
ds_raw.csv	Disposition / end of study	one row per subject	8
ex_raw.csv	Study drug exposure	one row per subject	8
ae_raw.csv	Adverse events	one row per event	9
cm_raw.csv	Concomitant meds	one row per medication	8
vs_raw.csv	Vital signs	WIDE — one row per visit	24
lb_raw.csv	Laboratory	tall — one row per test	48

Note vs_raw.csv is WIDE and lb_raw.csv is TALL — the same clinical idea, two different shapes. You'll handle both.

Everything is one of three types (plus missing)

Character

Text. Anything you don't do arithmetic on.

```
"ABC-01"  "F"  "bad headache"
```

IDs, codes, names, ISO dates

Numeric

Numbers you can calculate with.

```
50    36.7    122
```

Doses, results, ages

Date

A calendar point. Stored as a number internally, but SDTM wants ISO text.

```
2024-03-01
```

(character in SDTM!)

The fourth state: MISSING. A value that was not collected. In SAS a missing number is . and missing text is ""; in R both are NA. Missing is never the same as zero or an empty answer.

TRAP 1

Type guessing can destroy IDs

Readers guess column types from the values. An ID made only of digits looks like a number — and different tools make different guesses on the SAME file.

In the file

```
SITEID = 01  
SUBJID = 001
```

SAS PROC IMPORT base R read.csv()

```
SITEID = 1  
SUBJID = 1
```

readr read_csv()

```
SITEID = "01"  
SUBJID = "001"
```

→ If the zeros are lost, USUBJID becomes ABC-01-1-1 instead of ABC-01-01-001 — wrong for every subject, in every domain.

SAS

```
/* control the types yourself */  
data dm_raw;  
  infile "dm_raw.csv" dsd firstobs=2;  
  length siteid $3 subjid $5;  
  input studyid $ siteid $ subjid $ ...;  
run;
```

R

```
# declare intent, never rely on luck  
read_csv("dm_raw.csv",  
  col_types = cols(  
    SITEID = col_character(),  
    SUBJID = col_character(),  
    .default = col_guess()))
```

No raw date is ISO — convert every one

14-MAY-1969

DD-MMM-YYYY — the CRF standard

found in dm, ds, ex, vs, lb

15/03/2024

DD/MM/YYYY

found in ae, cm

10-Mar-2024

DD-Mon-YYYY

found in ae, cm

Two dangers. (1) 01/03/2024 is ambiguous — 1 March or 3 January? Check the data dictionary, never guess. (2) A blank date means ongoing or not done — leave it null, never substitute today's date. (3) DD-MMM-YYYY text does NOT sort chronologically — convert before any min/max.

Blank does not mean zero

1

Not collected

The test wasn't done at this visit.

HEIGHT is blank at BASELINE and WEEK 4 — height is measured once, at screening.

2

Still ongoing

The event or medication hasn't ended.

AEENDT is blank for the Nausea event — it was still ongoing at the end of the study.

3

Genuinely unknown

It was asked but the answer isn't known.

ETHNIC = 'Unknown' for one subject — that is a real recorded value, not a blank.

Different reasons, same appearance in the file. Understanding which one you're looking at is a clinical question, not a technical one.

Wide and tall are both “normal”

WIDE — vs_raw.csv

SUBJID	VISIT	SYSBP	DIABP	PULSE
001	SCREENING	122	80	68
001	BASELINE	120	78	66

One row per visit • one column per measurement

TALL — lb_raw.csv

SUBJID	VISIT	LBTEST	LBORRES
001	BASELINE	Hemoglobin	14.2
001	BASELINE	Hematocrit	42

One row per measurement • a test-name column

SDTM Findings domains are always TALL — so wide data must be transposed on the way in.

Recognising the shape is the first decision you make about a Findings file: does it need transposing, or is it already in the right form?

VS needs transposing (24 rows → 128).

LB does not (48 rows → 48).

The same idea, spelled many ways



Coded numbers

SEX = 1 / 2

You need the dictionary to know 1=Male, 2=Female



Inconsistent case

White / white / 'White '

Trim and upper-case before matching



Free text

'bad headache' / 'Headache'

Requires dictionary coding (MedDRA) — not a simple lookup



Synonyms

'No' / 'N'

Both mean the same; map both to CT 'N'

The inspection checklist

Run this on every unfamiliar dataset, every time. It takes two minutes and saves hours.

1

How many rows and columns?

Does the row count match what you expected?

2

What type is every column?

Especially the IDs — check the leading zeros survived.

3

How many distinct subjects?

Compare against DM. Any subject not in DM is a problem.

4

Which columns have blanks?

And what does each blank mean?

5

What are the distinct values?

For any column you'll map to CT — look at every value.

6

What date formats appear?

Check for more than one format in the same column.

Gap analysis: ABC-01 raw → SDTM

Inspection produces a to-do list. Here is the real one for our study.

What's wrong in the raw data	What you must do	Affects
SUBJID not unique study-wide	Build USUBJID	all
SEX stored as 1 / 2	Apply CT → M / F	DM
RACE free text, mixed case	Trim, upper-case, map to CT	DM
Birth date, no age	Derive AGE and AGEU	DM
Dates in 3 formats	Convert all to ISO 8601	AE, CM
No study day anywhere	Derive --DY from RFSTDTC	all
No record sequence	Derive --SEQ	all
VS is wide	Transpose to --TESTCD / --TEST	VS

Every one of these is documented in `data/mapping_specification.md` — your reference for the rest of the bootcamp.

AVOID THESE

Import mistakes that cost hours



Not checking the row count

If you imported 23 rows from a 24-row file, something was silently dropped.



Letting IDs become numbers

Leading zeros vanish and your USUBJID is wrong for every subject.



Assuming one date format

Check every date column for more than one pattern.



Treating blanks as zero

A missing weight is not a weight of 0 kg.



Editing the raw file

Never. Fix it in code so the step is documented and repeatable.



Skipping the dictionary

The data dictionary tells you what 1/2 and DD/MM mean. Read it first.

WHAT'S NEXT

Get the data in, then build your first domain

1

Notebook 03 Importing Raw Data — read all six files, run the checklist (SAS & R)

2

Module 05 Building the DM domain — Special Purpose, one row per subject

3

Notebook 04 Build DM end to end and check it against the reference dataset

You now have the finished answer in data/sdtm/ — but build it yourself first, then compare.